AN IMPROVED LOWER BOUND FOR IDENTIFYING CODES IN THE HEXAGONAL GRID VIA A STRICT LOCAL-TO-GLOBAL RELAXATION GAP

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Abstract. Every radius-one identifying code in the infinite hexagonal grid has lower asymptotic density at least

$$\frac{1427623}{3410330} = \frac{18}{43} + \frac{1}{79310},$$

improving Stolee's lower bound of 23/55. The value 18/43 is the exact optimum of a linear programming relaxation on a central face and its six neighbors, with the same one-face marginal distribution at all seven positions. The union of the supports of all optimal distributions consists of exactly 1,206 oriented stars. Each of these stars is locally admissible, yet no identifying code on the plane has its centered star in this union at every face. This rules out a plane realization of the optimal support but gives no numerical value for the density increment. To quantify the gap, we prove a separate obstruction for near-equality states. Every radius-15 ball in the face lattice contains a face whose final charge exceeds the certified lower bound on face charge by at least 1/55. Averaging over translates gives the stated increment for all identifying codes, including nonperiodic ones. The finite claims have fixed certificates that can be checked without re-solving the linear program or the SAT instances.

1. Introduction

How sparsely can one mark the vertices of the hexagonal grid so that the marks visible within one step identify every vertex? An identifying code, in the sense of Karpovsky, Chakrabarty and Levitin [1], requires each closed neighborhood to contain a mark and no two vertices to see exactly the same set of marks. For the infinite hexagonal grid, Cohen, Honkala, Lobstein and Zémor proved a lower bound of 16/39 and constructed a code of density 3/7 [2]. Cranston and Yu raised the lower bound to 12/29 [3], Cukierman and Yu to 5/12 [4], and Stolee to 23/55 by automated discharging [5, Theorem 1]. Salo and Törmä constructed a periodic code of density 53/126 [6, Theorem 1.1]. The exact density remains unknown.

Main result. Every radius-one identifying code C in the infinite hexagonal grid satisfies

$$\underline{\rho}(C) \;\geq\; \frac{18}{43} + \frac{1}{79310} = \frac{1427623}{3410330}.$$

Here the lower asymptotic density is taken over expanding boxes; the same bound holds over graph balls. No periodicity or existence of a limiting density is assumed. Relative to Stolee's 23/55 and the periodic upper bound 53/126 of Salo and Törmä, the new lower bound closes approximately 17.75% of the gap between them.

Why are there two terms in the bound? A configuration on the grid gives empirical frequencies of patterns around a face. Translating the observation window by one face changes these frequencies only through a boundary layer. We impose the resulting balance equations on a star consisting of a central face mask and its six neighbors. Within a star, the masks agree on every shared vertex; between stars, the linear program equates only their one-face marginal distributions. The resulting relaxation has exact minimum 18/43 (Theorem 3.1). These marginal constraints do not determine whether an optimal local distribution is induced by any identifying code on the plane.

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We determine the union of the supports of all optimal distributions. It contains exactly 1,206 oriented stars, but no identifying code on the plane has its star at every face in this set (Theorem 4.1). Each of these stars is locally admissible; the obstruction is to their global extension using only stars from the optimal support union. The distinction between locally consistent marginals and global realizability is familiar in dimension two [7]. The incompatibility result forces some stars outside the optimal support union. It does not assign a charge penalty to such a star and therefore gives no value for the density increment. To quantify the gap, we use the same discharging rule to prove a separate obstruction for stars with small central face surplus. Every radius-15 ball in the triangular lattice of face centers, which contains 721 faces, contains a face with surplus at least $1/55$ (Lemma 5.1). Each face is contained in exactly 721 such translated balls, and the asymptotic ratio of faces to vertices is $1/2$. Averaging over all translates gives the second term of the bound (Theorem 5.2).

For the local optimum we supply an exact feasible distribution and an integer discharging certificate. A separator and a second feasible distribution certify the full optimal support. For each of the two obstructions, the verifier reconstructs the overlaps in the original grid coordinates and checks a fixed elimination certificate without solving an optimization or SAT problem anew. The passage from the finite surplus obstruction to the lower bound for every plane code is analytic. The lower-bound proof uses the periodic construction at the upper endpoint only for comparison.

2. The hexagonal grid and the seven-face relaxation

We use brick coordinates for the hexagonal grid H. Its vertex set is $\mathbb{Z}^2$, and the neighbors of (x, y) are $(x-1, y)$, $(x+1, y)$, and $(x, y+(-1)^{x+y})$. Write $N[v]$ for the closed neighborhood of v. A set $C \subseteq V(H)$ is an identifying code if the sets $N[v] \cap C$ are nonempty and pairwise distinct.

Equivalently, C must meet every closed neighborhood and every symmetric difference $N[u] \triangle N[v]$ with $1 \le d_H(u, v) \le 2$. Vertices at greater distance have disjoint closed neighborhoods, so domination alone distinguishes them. Each required set has diameter at most four. Let $\xi : X \to \{0, 1\}$ be an assignment on a finite vertex set X and select every vertex outside X. The extension is an identifying code if and only if ξ meets every requirement supported entirely in X. Each other requirement contains a selected exterior vertex.

Translation cells and density. Let $e_1 = (2, 0)$ and $e_2 = (1, 1)$. The lattice $\Lambda = \mathbb{Z}e_1 + \mathbb{Z}e_2$ consists of the translations that preserve the parity of $x + y$. For $t \in \Lambda$, let $P_t = \{t, t + (1, 0)\}$. These two-vertex cells partition $V(H)$. Set

$$I_n = \{ie_1 + je_2 : -n \le i, j \le n\}, \qquad Q_n = \bigcup_{t \in I_n} P_t.$$

For a code C, define its lower asymptotic density by

$$\underline{\rho}(C) = \liminf_{n \to \infty} \frac{|C \cap Q_n|}{2(2n+1)^2}.$$

We write d_{hex} for the infimum of the limiting densities of identifying codes under this cell indexing. Salo and Törmä [6, Definition 3.1] take the limit of box averages when it exists and assign $+\infty$ to configurations without a limit when forming the infimum. Their Remark 3.2 concerns the value of that infimum under a change of averaging sets. Here $\underline{\rho}(C)$ is defined for each individual code, including codes whose density does not converge. Section 5 also proves the lower bound for each code along graph balls.

Faces, face masks, and stars. Let F denote the face with corners

$$(0,0),\ (1,0),\ (2,0),\ (2,1),\ (1,1),\ (0,1)$$

in cyclic order, and identify a face with its set of corners when taking unions. Every face is uniquely $F+t$ for some $t \in \Lambda$. Its six neighboring face offsets are

$$D = \{\pm e_1,\ \pm e_2,\ \pm(e_1 - e_2)\}.$$

Write $D_0 = D \cup \{0\}$. The *face mask* centered at F records the bits on $M = \bigcup_{a \in D_0}(F+a)$, with $|M| = 24$. A *seven-face star* records one face mask at each of the seven centers in D_0. Its site set is $G = \bigcup_{a \in D_0}(M+a)$, with $|G| = 54$ (Figure 1).



FIGURE 1. Supports used in the seven-face relaxation. Left: the central face F. Middle: the 24-site mask support M. Right: the 54-site support G of a seven-face star. Shading indicates the faces at which masks are read; it does not indicate selected code vertices.

A legal oriented star is a single assignment $s : G \to \{0,1\}$ satisfying every domination and separation requirement supported in G. Its mask at offset a is the word $t_a(s)$ on M given by $t_a(s)(p) = s(p+a)$. All seven words are read in the same coordinate frame; rotations and reflections are not identified. Thus every legal star is individually realizable in some identifying code on the plane. This does not imply that prescribed legal stars at different faces can be realized simultaneously. In a plane code, adjacent stars must agree on every vertex in their common support. The relaxation below replaces this condition between stars by equality of one-face marginal distributions.

Let $\mathcal{S}$ be the set of legal oriented stars, and let $k(s) = \sum_{p \in F} s(p)$. Let β_{star} denote the value of the following linear program:

$$\begin{aligned} \text{minimize} \quad & \frac{1}{6}\sum_{s \in \mathcal{S}} \mu_s k(s), \\ \text{subject to} \quad & \mu_s \geq 0, \qquad \sum_{s \in \mathcal{S}} \mu_s = 1, \\ & \sum_{s:\, t_a(s)=u} \mu_s = \sum_{s:\, t_0(s)=u} \mu_s \qquad (a \in D,\ u \in \{0,1\}^M). \end{aligned}$$

For a plane identifying code C, let s_t be the oriented star at $F+t$. Its empirical distribution over I_n is

$$\mu_n(s) = \frac{|\{t \in I_n : s_t = s\}|}{|I_n|}.$$

Choose a subsequence along which the density in Q_n tends to $\underline{\rho}(C)$. Since $\mathcal{S}$ is finite, pass to a further subsequence on which μ_n converges to a probability distribution μ on $\mathcal{S}$.

For any $a \in D$ the masks satisfy $t_a(s_t) = t_0(s_{t+a})$ in the original code. Hence for every mask word u,

$$\left| \sum_{s:t_a(s)=u} \mu_n(s) - \sum_{s:t_0(s)=u} \mu_n(s) \right| \le \frac{|I_n \triangle (I_n + a)|}{|I_n|}.$$

The numerator is $O(n)$ for each of the six fixed offsets. The limit distribution therefore satisfies all marginal equations.

To identify the objective, count the selected face corners before taking the limit. The three corners of F in Λ have offsets $0, e_1, e_2$; the other three have offsets $0, e_2, e_2 - e_1$ from $(1, 0)$. Thus the sum of the corner counts over I_n counts selected vertices in six translates of the corresponding parity classes of Q_n. Replacing each translated index set by I_n changes its count only on their symmetric difference, of size $O(n)$ independently of C. Each parity class then occurs three times, giving

$$\sum_{t \in I_n} k(s_t) = 3|C \cap Q_n| + O(n).$$

Since $|Q_n| = 2|I_n|$, the objective of μ_n is

$$\frac{1}{6} \sum_s \mu_n(s) k(s) = \frac{|C \cap Q_n|}{|Q_n|} + O\left(\frac{n}{|I_n|}\right).$$

Along the chosen subsequence the error tends to zero, so the feasible distribution μ has objective value $\underline{\rho}(C)$. In particular $\beta_{\text{star}} \le \underline{\rho}(C)$ for every plane identifying code. Section 4 shows why the reverse realization need not hold.

3. Exact solution of the local relaxation

Theorem 3.1. The seven-face relaxation has exact minimum $\beta_{\text{star}} = 18/43$.

Feasible distribution (upper bound on the LP minimum). The primal certificate gives 563 legal stars with positive integer weights of total 774. All seven marginal histograms agree exactly. The weighted central corner count is 1,944. Hence this distribution attains

$$\frac{1944}{6 \cdot 774} = \frac{18}{43}.$$

Discharging certificate (lower bound). Fix a legal star s. Put initial charge $s(v)$ on each vertex $v \in G$ and zero on every face. Transfers are signed, so a negative transfer from a vertex to a face means a transfer in the opposite direction. Put $d = 66220$ and measure charge in units of $1/(165d)$. For each incidence needed to compute the retained charges at the six corners and the charge of the central face, the transfer from v to f is

$$d(55s(v) - 23) + z_{i(v,f)}$$

integer units. The directed word $i(v, f)$ records the 13 sites of the three faces incident to v, with v as its low bit. Starting with f at v, read its corners cyclically, then those of the remaining incident faces in cyclic order, omitting repeated sites. This is the neighborhood information of Stolee's rule N [5]. Subtracting the three transfers, the vertex retains physical charge

$$r_s(v) = \frac{23}{55} - \frac{1}{165d} \sum_{f \ni v} z_{i(v,f)}.$$

The verifier checks $\sum_{f \ni v} z_{i(v,f)} \le 0$ on every legal vertex neighborhood.

Across an edge the net charge received by f, in integer units, is

$$\phi(b) + m(t) - m(u).$$

Here b is the directed 16-site overlap word and t, u are the two 24-site masks oriented in their respective edge frames. Reversing the edge negates ϕ and interchanges the requests $m(t), m(u)$. The tables are part of the discharging certificate; see Section 6 for the verification procedure.

Computational Lemma 3.2. For every legal star s, if $q_s(F)$ denotes the final charge at its central face in physical units, then its integer balance $Q_s(F) = 165d\, q_s(F)$ satisfies $Q_s(F) \geq 9240$.

Proof. Let W be the rotation-closed request list. All requests are nonnegative, and each nonzero request is supported on W. Fix a legal central mask u. Its corner transfers, the six ϕ terms and its own requests are determined by u. The only remaining terms in its balance are the requests paid to its six neighbors.

If $u \notin W$, its own requests are zero. In each direction the verifier reads the shared word in the neighbor's edge frame. The envelope for that word bounds every request of a neighbor with that overlap: the table is checked on every request-list mask, and requests off the list vanish. Subtracting the sum of the six envelopes gives a lower bound for $Q_s(F)$ for **every** legal completion of u, even if the six envelope values cannot be attained together.

If $u \in W$, the verifier places each candidate neighbor with a positive request on its original sites. It keeps choices that agree with u and with each other wherever their sites overlap. In each direction it also allows a choice of zero with no additional bits fixed. The positive requests in any legal completion form one of the choices in this maximization, so the maximum bounds that completion's total expenditure from above. Subtract it from the balance fixed by u.

The smallest certified lower balance in each class is 9,240 integer units. Thus every legal star satisfies $q_s(F) \geq 9240/(165d)$. □

The certificate is used only to establish the following pointwise bounds:

$$r_s(v) \geq \frac{23}{55} \qquad (v \in F), \qquad q_s(F) \geq \delta, \qquad \delta = \frac{9240}{165d},$$

for every legal star s. The remainder of the lower-bound argument is an exact expectation identity valid for every feasible distribution.

Lemma 3.3. For every feasible distribution μ, the pooled charge

$$P(s) = q_s(F) + \frac{1}{3}\sum_{v \in F} r_s(v)$$

satisfies $\mathbb{E}_\mu P(s) = \mathbb{E}_\mu k(s)/3$.

Proof. For every $a \in D_0$ and every test function $h : \{0,1\}^M \to \mathbb{R}$, the one-face marginal equations give

$$\mathbb{E}_\mu h(t_a(s)) = \mathbb{E}_\mu h(t_0(s)).$$

We apply this identity to individual transfer terms after recentering their masks.

Write $T(v, g)$ for the transfer in physical units from v to g. When v is a corner of F and g is incident to v, the mask at g determines $T(v, g)$. Translating g to F does not change this expectation, by the marginal equations. As (v, g) ranges over these incidences, each corner of the recentered face occurs three times, once for each original face incident to that corner. Thus

$$\frac{1}{3}\sum_{v \in F}\sum_{g \ni v} \mathbb{E}_\mu T(v, g) = \sum_{u \in F} \mathbb{E}_\mu T(u, F).$$

The debits at the vertices therefore cancel the central face's receipts in expectation.

Let J_a be the signed receipt at F across its edge in direction a, in integer units. Write m_a for the request of a face mask read in that edge frame. The request part of J_a is $m_a(t_0) - m_{-a}(t_a)$. Its expectation is therefore $\mathbb{E}_\mu[m_a(t_0) - m_{-a}(t_0)]$, which reverses sign for $-a$. For the shared-word part, the neighbor reads the same bits in the opposite frame, where ϕ has the opposite sign. By the test-function identity, this reversed contribution read at t_a has the same expectation as the contribution read at the central mask in direction $-a$. Consequently

$$\mathbb{E}_\mu[J_a + J_{-a}] = 0$$

for each opposite pair. This cancellation uses only single-mask marginals.

Only the initial charges of the six corners remain, with coefficient 1/3. This gives the asserted identity. □

Proof of Theorem 3.1. Let μ be any feasible distribution. By the pointwise floors and Lemma 3.3,

$$2\left(\frac{1}{6}\mathbb{E}_\mu k(s)\right) = \mathbb{E}_\mu P(s) \geq 2\frac{23}{55} + \delta.$$

Therefore every feasible distribution satisfies

$$\frac{1}{6}\mathbb{E}_\mu k(s) \geq \frac{23}{55} + \frac{\delta}{2} = \frac{23}{55} + \frac{1}{2}\frac{9240}{165 \cdot 66220} = \frac{18}{43}.$$

Every feasible distribution thus has objective value at least 18/43, and the certified feasible distribution has value 18/43. Therefore $\beta_{\text{star}} = 18/43$. □

4. The optimal support union cannot cover the plane

The 563 stars in the primal certificate exhibit one optimal distribution. Other optimal distributions may use different stars. Let

$$U = \{s \in \mathcal{S} : \mu_s > 0 \text{ for some optimal feasible distribution } \mu\}.$$

We determine U and ask whether an identifying code on the plane can have all its stars in U.

Overlap compatibility. Fix $a \in D$. Two oriented stars s, s' are compatible across a if placing s at the origin and s' at a gives the same bit at every site covered by both:

$$s(p) = s'(p - a) \qquad \text{for every } p \in G \cap (G + a).$$

The comparison uses the full intersection in the original grid coordinates. In any plane code, neighboring stars satisfy this condition in each of the six directions.

Mask pruning preserves the optimal marginal support (auxiliary lemma). Let A be the support of the common one-face marginal of any optimal feasible distribution. At every round of the equality-mask pruning, the surviving set contains A.

Proof. Fix an optimal distribution μ. By Lemma 3.3, its objective value 18/43, and $2(18/43) = 2(23/55) + \delta$, we have

$$\mathbb{E}_\mu\left[q_s(F) - \delta + \frac{1}{3}\sum_{v \in F}\left(r_s(v) - \frac{23}{55}\right)\right] = 0.$$

Each contribution is nonnegative. Hence at every positive-mass star the central face charge is exactly δ and all six retained vertex charges are exactly 23/55.

For $u \in A$, choose a positive-mass star s with $t_0(s) = u$. Request-list masks are retained unconditionally. For any other u, write $\ell(u)$ for its envelope lower bound on the **total** central face charge in integer units (Section 3). Then

$$\ell(u) \leq 165d\, q_s(F) = 165d\,\delta = 9240 < 9240 + 1.$$

Thus u passes the initial equality-mask test. The cutoff of one integer unit selects every equality case, including cases in which the envelope underestimates the actual charge.

Suppose a current list contains A. Each of the six other masks in the chosen star has positive mass in the marginal at its offset, so belongs to A by marginal balance. Each agrees with u on the corresponding overlap. It witnesses survival of u in that direction. This argument applies to each $u \in A$ at every round. All witnesses used here are masks within a single star of the distribution. □

Theorem 4.1 (Optimal stars and planar incompatibility). Exactly 1,206 oriented stars belong to U. No identifying code on the plane has all its stars in U.

Proof. For the equality test the successive mask-list sizes are

$$65496, \quad 768, \quad 516, \quad 402, \quad 378.$$

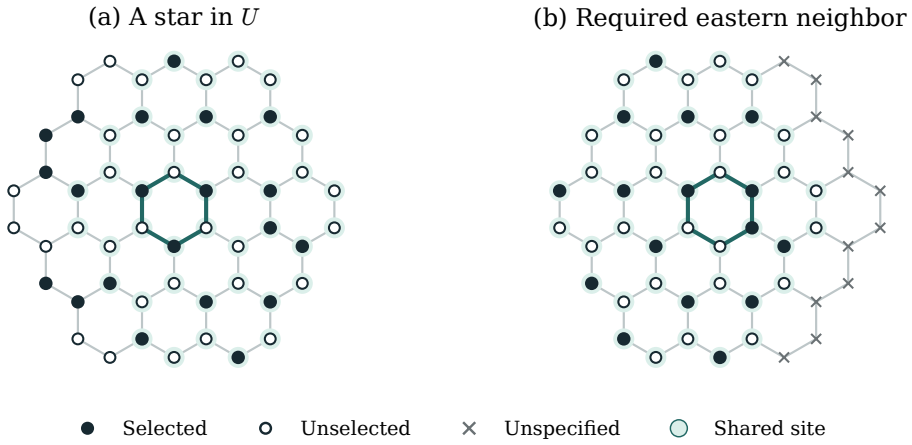


FIGURE 2. A first-round deletion in the eastward direction $a = (2, 0)$. Panel (a) shows an oriented star in U. In panel (b) the 42 shared sites inherit their bits from (a); crosses mark sites whose bits are still free. Among all 1,206 stars in U, none agrees with the 42 prescribed bits in the eastern frame. The outlined hexagon marks the central face in each panel.

Join seven surviving masks on their shared sites, retain zero central surplus, and check the original identification requirements and all six corner equalities. By the auxiliary lemma, all seven masks of every positive-mass star of every optimal distribution survive. The join reconstructs their original bits. Hence the resulting list of 1,914 candidates contains U.

Two integer certificates determine which candidates can carry mass. For the first, let y be the integer table in the support certificate and R run over the six rotations about the face center. The functional

$$L(s) = \sum_{R} \sum_{a \in D} \left(y(a, t_0(Rs)) - y(a, t_a(Rs)) \right)$$

has expectation zero under every feasible distribution. To see this, let R act on a word by $(Rs)(p) = s(R^{-1}p)$, and on a mask in the same way. On translation offsets write Ra for the displacement taking $R(F)$ to $R(F + a)$. For each fixed R and a the mask definition then gives

$$t_0(Rs) = R\, t_0(s), \qquad t_a(Rs) = R\, t_{R^{-1}a}(s).$$

Apply the marginal equation in direction $R^{-1}a$ to the test function $u \mapsto y(a, Ru)$. The two terms for this R, a have the same expectation under μ. Summing gives $\mathbb{E}_\mu L(s) = 0$ without any rotation-invariance assumption on μ.

Let U^* denote the zero set of L **within the 1,914 candidates**. The separator is nonnegative there and strictly positive on 708 candidates. Every optimal distribution is supported on these candidates, so its zero expectation forces its support into U^*. This proves $U \subseteq U^*$.

Conversely, the second certificate gives strictly positive integer weights on all 1,206 members of U^* and zero elsewhere. Their sum is 20,124. After normalization the seven marginal histograms agree and the objective is $18/43$. This optimal feasible distribution witnesses $U^* \subseteq U$. Hence $U = U^*$ and $|U| = 1{,}206$.

Global nonextendability of U. This test compares two whole stars at adjacent faces. Agreement on their full shared support is necessary in a plane code, whereas the LP imposes only equality of one-face marginals. Set $U_0 = U$ and define

$$U_{j+1} = \{ s \in U_j : \forall a \in D,\ \exists s' \in U_j \colon s' \text{ is compatible with } s \text{ across } a \}.$$

If a plane code used only stars in U_0, every star occurring in it would belong to every U_j: its six actual neighbors witness survival in the next round. The verifier reconstructs these lists

from the shared coordinates without identifying reflected states. The successive cardinalities are

$$1206 \longrightarrow 726 \longrightarrow 480 \longrightarrow 264 \longrightarrow 126 \longrightarrow 36 \longrightarrow 18 \longrightarrow 0.$$

Figure 2 shows one of the stars eliminated in the first round. Thus no such plane code exists. □

Every star in U can occur in some plane code by selecting the exterior. What fails is the requirement that a single plane code have its star in U at every face. Exclusion from U does not by itself certify positive face surplus: membership in U also depends on marginal balance and vertex charges. The deletion argument alone therefore gives no numerical density increment. We next test the face surplus itself, allowing stars outside U.

5. A quantitative local-to-global gap

Let $c(F+t)$ be the final charge at face $F+t$ under the discharging rule of Section 3, and write

$$\sigma(t) = c(F+t) - \frac{9240}{165d}, \qquad d = 66220.$$

Thus $\sigma(t) \geq 0$ in every identifying code. We now allow every local configuration with $\sigma(t) < 1/55$, including configurations outside the optimal support of Section 4. Let $B_r(t)$ denote the radius-r ball about t in the face lattice, with adjacency given by D.

Lemma 5.1 (Radius-15 surplus obstruction). For every identifying code and every face center t,

$$\max_{u \in B_{15}(t)} \sigma(u) \;\geq\; \frac{1}{55}.$$

Proof. Suppose $\sigma(x) < 1/55$ for all $x \in B_{15}(t)$. Let W be the request list of Section 3. For a legal central mask $u \notin W$, let $\ell(u)$ be the central charge estimate in integer units obtained by subtracting the six outgoing request envelopes from the charge fixed by u. Each envelope bounds the corresponding expenditure from above. Hence every legal completion ω of u satisfies $\ell(u) \leq 165d\, c_\omega(F)$, where $c_\omega(F)$ is the final face charge in physical units, with the face recentered at F.

Define $\mathcal{A}_0$ to contain W and all legal $u \notin W$ with $\ell(u) < 9240 + 198660$. If u occurs at $x \in B_{15}(t)$ in the assumed code and $u \notin W$, its actual completion gives

$$\ell(u) \;\leq\; 165d\, c(F+x) = 9240 + 165d\, \sigma(x) < 9240 + 198660,$$

since $165d/55 = 3d = 198660$. Masks in W are retained unconditionally. Thus every actual mask in $B_{15}(t)$ belongs to $\mathcal{A}_0$.

Six mask rounds. A mask in $\mathcal{A}_i$ survives in $\mathcal{A}_{i+1}$ if it has an agreeing neighbor in $\mathcal{A}_i$ in each of the six directions, on the intersection of the two mask supports. If every actual mask in $B_{15-i}(t)$ belongs to $\mathcal{A}_i$, the six neighbors of any face in $B_{14-i}(t)$ witness its survival. Induction for $i = 0, \ldots, 5$ leaves every actual mask in $B_9(t)$ on the list. The list sizes, including the initial one, are $141780, 1578, 852, 624, 552, 540, 534$.

One star join. Center a star in $B_8(t)$. All seven of its mask centers lie in $B_9(t)$, so their masks survive and agree on the original grid sites. Join all consistent seven-tuples of surviving masks and evaluate the central face charge using the seven actual mask words of each tuple. Keep the joined words below the strict cutoff. The resulting set $\mathcal{T}_0$ has 5,748 elements and contains every actual star in $B_8(t)$. The join may also retain words which cannot occur there. This list is defined by face charge, independently of the marginal equations used to determine U.

Eight star rounds. Apply the same direction-by-direction test to $\mathcal{T}_0$, this time comparing whole 54-site stars on their full translated intersections. If the actual stars throughout $B_{8-j}(t)$ survive round j, their neighbors witness survival throughout $B_{7-j}(t)$ at round $j+1$. The certificate assigns to each deleted star a direction and a deletion depth at most eight. Every

candidate agreeing in that direction has smaller depth, with no agreeing candidates at depth one. Thus $\mathcal{T}_8$ is empty. The induction would put the actual central star in $\mathcal{T}_8$, giving the contradiction. For the Boolean certificate, set X_s true exactly when $s \in \mathcal{T}_0$ occurs somewhere in $B_{8-\mathrm{depth}(s)}(t)$. The center clause is

$$\bigvee_{s \in \mathcal{T}_0} X_s.$$

Whatever its deletion depth, the actual star at t makes its own variable true because its occurrence ball contains t. If X_s is true at depth i, choose a witnessing occurrence and take its actual neighbor in the direction specified for s. This neighbor lies in $B_{8-i+1}(t) \subseteq B_8(t)$, so its state s' belongs to $\mathcal{T}_0$. It agrees with s in the specified direction and has depth $i' < i$. Since $8 - i + 1 \leq 8 - i'$, the same occurrence witnesses $X_{s'}$. The geometric hypothesis would therefore satisfy the center clause and each direction implication in the certificate. Section 6 verifies their inconsistency. □

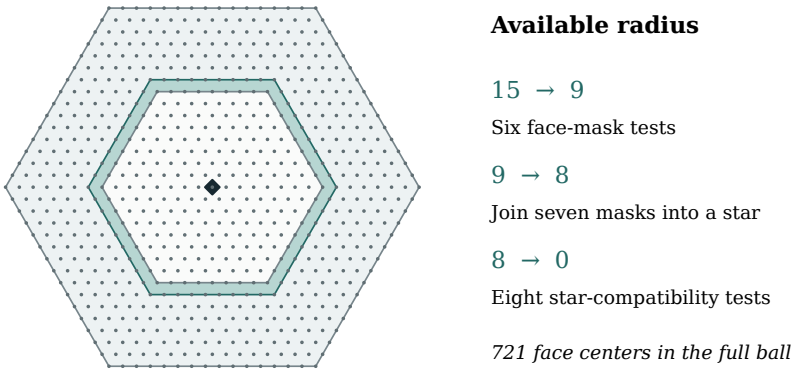


FIGURE 3. The radius-15 ball on the triangular lattice of face centers. Outlines mark radii 15, 9 and 8; the diamond marks the center. By symmetry the 721 points are also the possible centers of radius-15 translates containing the diamond. Restricting to interior translates can only reduce this multiplicity in (1).

Theorem 5.2. Every identifying code C in the hexagonal grid satisfies

$$\underline{\rho}(C) \;\geq\; \frac{18}{43} + \frac{1}{79310}.$$

Proof. A ball of radius 15 in the triangular face lattice contains $1 + 6\sum_{j=1}^{15} j = 721$ faces. Use the index boxes I_n and vertex boxes Q_n of Section 2. For $n \geq 15$, every $t \in I_{n-15}$ satisfies $B_{15}(t) \subseteq I_n$. Lemma 5.1 and the nonnegativity of all face surpluses give

$$\frac{|I_{n-15}|}{55} \;\leq\; \sum_{t \in I_{n-15}} \sum_{u \in B_{15}(t)} \sigma(u) \;\leq\; 721 \sum_{u \in I_n} \sigma(u). \tag{1}$$

The second inequality counts each face u at most 721 times, once for each possible center of a ball containing it.

There are exactly $|I_n|$ indexed faces and $2|I_n|$ vertices in the corresponding cells. On summing the discharging identities, internal transfers cancel. The remaining transfers cross a boundary of fixed thickness. Their total absolute value is at most Kn for a constant K independent of the code and of the translate of the box. Indeed, let M bound the absolute physical transfer on any one incidence or face edge in the fixed tables. A transfer that does not cancel has one endpoint inside and one outside the selected vertex and face regions. These endpoints lie in a collar of fixed thickness in cell coordinates. The collar has at most cn incidences and face edges

for $n \geq 15$, with c depending only on the cell geometry. We may take $K = cM$. The vertex charge floor and the face charge floor together contribute the baseline $18/43$ per vertex. Hence

$$|C \cap Q_n| \;\geq\; \frac{18}{43}|Q_n| + \sum_{u \in I_n} \sigma(u) - Kn. \tag{2}$$

Combining (1) and (2) gives the finite-box estimate

$$\frac{|C \cap Q_n|}{|Q_n|} \;\geq\; \frac{18}{43} + \frac{|I_{n-15}|}{|I_n|}\,\frac{1}{55 \cdot 721 \cdot 2} - \frac{Kn}{2|I_n|}.$$

The three factors come from the forced surplus of $1/55$, the covering multiplicity of at most 721, and the ratio of one face to two vertices. Here $|I_{n-15}|/|I_n| = (2n-29)^2/(2n+1)^2 = 1 - O(1/n)$ and $Kn/(2|I_n|) = O(1/n)$, with both error bounds uniform in the translate and the code. Taking a lower limit yields the asserted increment $1/79310$.

For graph balls, fix $k \geq 15$ and tile the plane by disjoint cell boxes of the form $Q_k + t$. Let V_R be a vertex ball of radius R. Retain every box wholly contained in V_R. The uncovered vertices of the ball lie in boxes meeting both V_R and its complement. Each such box has graph diameter $O(k)$, so its vertices inside the ball lie in a shell of width $O(k)$, containing $O(kR)$ vertices for R large relative to k. To see the shell estimate, note that on either parity class the two-step neighbors form a triangular lattice: its rings have $6j$ vertices at radius $j \geq 1$, and vertices of the other parity are one step from this class. The ball has $\Theta(R^2)$ vertices. Thus $|V_{R+1}|/|V_R| \to 1$, the slow growth property of Sampaio, Sobral and Wakabayashi [9, Definition 3.1]. Their Lemma 3.2(a) gives the same limit for any fixed radius increment. Here the shell count also bounds the loss for boxes of diameter $O(k)$. In each retained box the loss from the interior ratio and the boundary flux is $O(k)$ in the count of code vertices, uniformly over translates. There are $O(R^2/k^2)$ retained boxes. Writing $b = 18/43 + 1/79310$, we obtain, for $15 \leq k$ and R sufficiently large compared with k,

$$\frac{|C \cap V_R|}{|V_R|} \geq b - \frac{A}{k} - \frac{Bk}{R}.$$

The first error bounds the sum of all per-box losses, and the second bounds the discarded shell. Both constants depend only on the grid and the fixed transfer tables. Taking $\liminf_{R\to\infty}$ for each fixed k gives a lower bound of $b - A/k$. Taking $k \to \infty$ gives b for every code and every choice of ball center. □

In particular, taking the infimum over codes with a limiting density gives

$$d_{\mathrm{hex}} \geq \frac{18}{43} + \frac{1}{79310}.$$

6. Certificates and reproducibility

The verifier checks four finite premises using three families of certificates. Sections 3–5 prove the conservation identity, the support inclusions, the geometric inductions and the passage to infinite-plane density.

Local optimum. The first family supplies a feasible distribution of value $18/43$ supported on 563 stars and an integer discharging rule. The verifier checks the marginal histograms, the objective value and the pointwise charge inequalities. The expectation identity in Lemma 3.3 converts these inequalities into a lower bound on the objective of *every* feasible distribution. Together these facts give the exact optimum.

Support of the optimal distributions. The second family concerns the 1,914 candidates generated by the equality test of Section 4. The verifier checks that the separator is nonnegative on this list, positive on 708 stars, and zero on precisely the support of a second feasible optimal distribution with 1,206 positive weights. The proof in Section 4 explains why the list contains every optimal support and why the separator has zero expectation. These arguments and the two finite checks give the two inclusions in Theorem 4.1.

Two overlap obstructions. The third family has two separate inputs and traces.

- For the optimal-support obstruction, start with the 1,206 stars of U. The verifier reconstructs the full overlaps and checks a deletion certificate of depth seven. Any actual configuration using only U on a radius-seven face ball would survive at its center through all seven rounds, by the induction of Section 4.
- For the surplus obstruction, start with the legal face alphabet. The verifier rebuilds the low-surplus mask list, applies six mask-deletion rounds, joins seven masks and tests the strict charge cutoff. It compares the resulting 5,748 stars with the stored list before checking a full-star deletion certificate of depth eight. The corresponding Boolean trace covers these last eight layers. The preceding list construction is regenerated from the coordinates and tables. The induction of Lemma 5.1 shows why each stage retains the hypothesized low-surplus configuration in the required inner ball.

Meaning of the Boolean traces. Let S be the initial list for the *full-star* deletion stage and let K be its certified depth (seven or eight, respectively). Under the corresponding geometric hypothesis every actual star centered in $B_K(t)$ belongs to S. For a state s of depth $i(s)$, assign X_s true exactly when s occurs somewhere in $B_{K-i(s)}(t)$. In the designated direction $a = a(s)$ the checker reconstructs all neighbors $N_a(s)$ compatible on the shared grid sites. They have strictly smaller depth. An occurrence witnessing X_s has an actual neighbor in $B_{K-i(s)+1}(t)$; that occurrence lies within the allowed radius of its own state. The assignment therefore satisfies the implication for each s and also the center clause:

$$\neg X_s \ \vee \bigvee_{s' \in N_a(s)} X_{s'}, \qquad \bigvee_{s \in S} X_s.$$

The actual center makes its own variable true because t belongs to every occurrence ball. At depth one the neighbor list is empty, forcing X_s false. Unit propagation in increasing depth forces all variables false, contradicting the last clause. Each supplied DRUP trace [8] thus adds only the empty clause. The verifier checks the clause lists against the original coordinates and checks this propagation order. The proof needs only the implication from a geometric configuration to a Boolean assignment.

Arithmetic and replay. From the extracted certificate directory run `python verify.py` using 64-bit Python with NumPy and assertions enabled. The verifier first compares the listed files with their SHA-256 digests in `CERTIFICATE_MANIFEST.json`. The manifest establishes consistency with the specified files; identifying the authoritative manifest requires a separately authenticated release.

The checks use Python integers and exact fractions, together with fixed-width NumPy integer operations. A face mask occupies 24 bits, a directed overlap 16 bits, and an assembled star 54 bits. The enumeration uses signed 32-bit arrays for vertex and face words, and signed 64-bit arrays for the stored star words and charge tables. Before the vectorized computation, `check_ranges.py` bounds the fixed coefficients and charge subtotals using Python integers. With $d = 66220$, an incidence has absolute value at most $32d + \max|z|$. A face has six incidences, six overlap terms and six pairs of requests. For the fixed tables the resulting absolute bound, including subtraction of the charge floor, is 52,499,184. The preflight also bounds the envelope case, the archived calibration sums and the potential differences for the supplementary strip checks. The tested versions, resource measurements and expected terminal output are recorded in `REPLAY_NOTES.txt`.

A successful run verifies the finite data and inequalities against the supplied verifier, interpreter, NumPy implementation and arithmetic bounds. Sections 3–5 supply the mathematical implications, including the averaging and boundary limits for nonperiodic codes. Replaying the package does not require an optimization or SAT search.

7. Discussion

The seven-face relaxation has exact minimum 18/43. The union of the supports of its optimal distributions consists of 1,206 oriented stars. Each of these stars extends individually to a plane identifying code, but no plane identifying code has its centered star in this union at every face.

The explicit density increment comes from a separate near-equality obstruction. Every radius-15 ball of face centers contains a face with surplus at least 1/55. Each face lies in exactly 721 translates of this ball. Together with the ratio of one face to two vertices, this gives an increment of 1/79310 above the relaxation optimum for the lower asymptotic density of every code, including nonperiodic codes.

Together with the periodic construction of Salo and Törmä [6], this gives

$$\frac{1427623}{3410330} \leq d_{\mathrm{hex}} \leq \frac{53}{126}.$$

The remaining question is the exact value of d_{hex}. One route to a sharper lower bound is to determine how much face surplus a plane code must carry on average. Our estimate counts one face of surplus at least 1/55 per radius-15 ball and uses only nonnegativity for the other faces.

Use of computational and language tools. OpenAI tools were used to assist with computational exploration, SAT and linear-programming formulations, software implementation, and manuscript preparation.

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